TQUK Functional Skills Qualification in Maths at Level 2

Examination Paper (Sample Assessment Paper 2)

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

Information:

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

SAMPLE

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

- Assessment: Sample Paper 2 (Non-Calculator & Calculator)
- Complete stepped calculations, methods & mark scheme codes [M1/A1]
- Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)
- Pass standard: 35 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Section B	45		
Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 0.872 + 0.15

Calculate: 6.672 – 5.65 = 1.022

Answer	1.022 ✓ [1 mark]

[1]

2. Work out 2.205 × 4

Calculate: 1.8 × 4.9 = 8.82

Answer	8.82 ✓ [1 mark]

[1]

SAMPLE

Questions continue on the following page

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3. A cyclist cycles 3 miles in 10 minutes.

What is the cyclist's average speed in miles per hour (mph)?

Speed = 3 miles in 10 mins = (3 ÷ 10) × 60 = 18 mph

Answer	18 ✓ mph

[2 marks]

[2]

4. Write 0.86:

(a) as a fraction

(b) as a percentage.

Decimal 0.86: Fraction = 86/100 (or 43/50) | Percentage = 86%

Answer	**(a) Fraction:** 86/100	**(b) Percentage:** 86% ✓

[2 marks]

[2]

SAMPLE

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5. Jordan goes out for lunch.

The bill comes to £20

Jordan wants to leave extra money as a tip.

The extra money is 18% more than the bill.

What is the total amount that Jordan will pay?

VAT at 18%: £20 × 1.18 = £23.60

Answer	£ 23.60 ✓ [2 marks]

[2]

SAMPLE

Questions continue on the following page

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6. Work out the median of these numbers:

5000 3000 2500 4250 5500 6600

Ordered numbers: 2500, 3000, 4250, 5000, 5500, 6600

Median = (4250 + 5000) ÷ 2 = 9250 ÷ 2 = 4625

Answer	4625 ✓ [2 marks]

[2]

SAMPLE

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7. A map has a scale of 1 : 50 000

A school and a supermarket are 2.5 centimetres (cm) apart on the map.

How far apart are the school and supermarket in real life?

Give the units in your answer.

Map scale: 2.5 cm × 50 000 = 125 000 cm = 1.25 km (1250 m)

Answer 1.25 km (1250 m) [2 marks]

[2]

SAMPLE

Questions continue on the following page

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8. A company gives its staff 6 pizzas to share.

Each pizza is cut into 8 slices.

On the first day, the staff eat 3 whole pizzas and 5 additional slices.

On the next day, they eat another 10 slices of pizza.

How much pizza is left after the second day?

Give your answer as a mixed number.

Common denominator: 3 5/8 + 1 2/8 = 4 7/8

Remaining = 6 – 4 7/8 = 5 8/8 – 4 7/8 = 1 1/8

Answer	1 1/8 ✓	[3 marks]

[3]

SAMPLE

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End of Section A.

SAMPLE

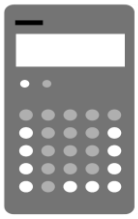
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Section B begins on Page 8.

SAMPLE

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Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. Put these numbers in order starting with the lowest:

–865 –811 –820 –814

Show any working here:

Answer	-865 , -820 , -814 , -811		✓	[1 mark]
	Lowest	→		**Highest**

[1]

2. There are 2.54 centimetres (cm) in 1 inch.

How many inches are there in 127 cm?

Conversion from graph: 127 cm = 50 inches

Answer	50 ✓	[1 mark]

[1]

Questions continue on the following page

SAMPLE

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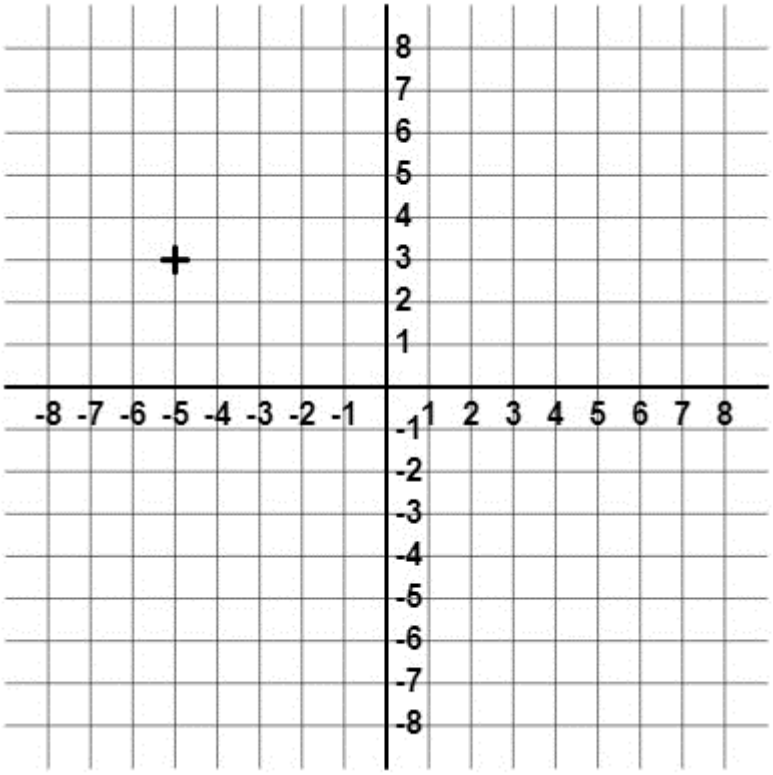
3. Calculate $12^2 - 3 \times 2.1$

BIDMAS calculation: 137.7

Answer	137.7 ✓ [1 mark]

[1]

4. Write down the coordinates of the point labelled on the grid.



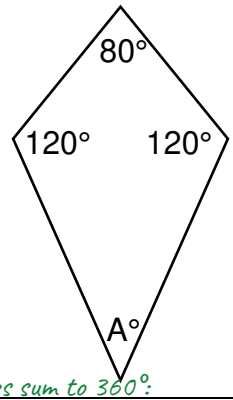
Answer	(-5, 3) ✓ (,) [1 mark]

[1]

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5. Work out the value of angle A.

Not drawn to scale



Quadrilateral interior angles sum to 360°:

$360° - 120° - 120° - 80° = 40°$

Answer 40 ✓ [2 marks]°

[2]

SAMPLE

Questions continue on the following page

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6. A garden has a circular flowerbed in the centre.

The diameter of the flowerbed is 1.38 metres (m).

Children make fairy lights to go all around the edge of the flowerbed.

The total length of the fairy lights is 4.5 metres (m).

Are the fairy lights long enough to decorate all around the edge of the flowerbed?

Show how you decide.

Use $\pi = 3.14$

Circumference = π × d = 3.14 × 1.38 m = 4.3332 m

Comparison: 4.3332 m < 4.5 m. Ribbon is long enough. Yes.

Answer	Yes (4.33 m < 4.5 ✓) [3 marks]

[3]

SAMPLE

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7. Charlie recently used 14 litres of paint to decorate the living room walls.

The total area painted was 91 m^2

Charlie now wants to paint the bedroom walls.

The total area to be painted is 78 m^2

Charlie has 11 litres of paint.

Does Charlie have enough paint?

Show how you decide.

Coverage rate = 91 m^2 ÷ 14 litres = 6.5 m^2 per litre

Area covered by 11 litres = 11 × 6.5 = 71.5 m^2

Comparison: 71.5 m^2 < 78 m^2 (needs 12 litres). Therefore No.

Answer	No (71.5 m^2 < 78 m^2) ✓ *[3 marks]*

[3]

SAMPLE

Questions continue on the following page

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8. Morgan is planning a holiday and finds this information:

Greenland	Average temperature 14° Fahrenheit (°F)
Iceland	Average temperature –11° Celsius (°C)

Morgan wants to go to the warmer country.

Which country should Morgan go to?

Show how you decide.

Use: $C = \frac{5(F-32)}{9}$

Where: C = temperature in degrees Celsius
F = temperature in degree Fahrenheit

Formula: C = 5(F - 32) ÷ 9 --> C = 5(14 - 32) ÷ 9 = 5(-18) ÷ 9 = -10°C

Comparison: -10°C < 0°C. Greenland is colder than Iceland.

Answer	Greenland (-10°C < ✓ [3 marks]

[3]

SAMPLE

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9. This year, Jo's gym membership price has increased by 5%.

The new monthly price is £33.60

How much did Jo pay for an entire year's membership last year?

Original monthly price: £33.60 ÷ 1.05 = £32.00

Annual cost before increase = £32 × 12 = £384

Answer	£ 384 ✓

[3 marks]

[3]

SAMPLE

Questions continue on the following page

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10. An athletics club awards points each year to its members.

In 2021, the club awarded its members a mean of 90.3 points in the year.

The grouped frequency table shows the points the club awarded its members in 2022:

Number of points awarded	Frequency
0 ≤ points < 40	7
40 ≤ points < 80	34
80 ≤ points < 120	100
120 ≤ points < 160	9
Total	**150**

The club president says:

'The mean number of points per member in 2022 has increased since 2021'

Is the president correct?

Show how you decide.

Midpoints: 20, 60, 100, 140. Total frequency = 7 + 34 + 100 + 9 = 150

Total fx = (20×7) + (60×34) + (100×100) + (140×9) = 140 + 2040 + 10 000 + 1260 = 13 440

Estimated Mean = 13 440 ÷ 150 = 89.6 (not 90). Therefore No.

Answer No (Mean is 89.6, not 90) [4 marks]

[4]

SAMPLE

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11. Hayden is organising activities for 80 students going on a school trip.

The ratio of the number of students who can participate in swimming, rock climbing and go karting is 5 : 3 : 2

Complete the activity list for Hayden.

Activity	Number of Students
Swimming	
Rock climbing	
Go karting	
Total	

Ratio Swimming : Climbing : Karting = 5 : 3 : 2 --> Total = 10 parts

1 part = 80 ÷ 10 = 8 students

Swimming = 5 × 8 = 40 | Climbing = 3 × 8 = 24 | Karting = 2 × 8 = 16

Show any workings here:

40

24

16

[4 marks]

[4]

SAMPLE

Questions continue on the following page

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12. Harper is a manager of a company.

Last year the company made a profit of nine hundred and seventy-five thousand and fifty pounds.

This year Harper wants to increase the profit by 3.5%.

Harper thinks the new profit will be more than £1 010 000

Is Harper correct?

Show how you decide.

House price = £975 050. Value after 3.5% increase: £975 050 × 1.035

New Value = £1 009 176.75

Comparison: £1 009 176.75 < £1 010 000. Target not reached. No.

Answer	No (£1 009 177 < £1 [4 marks]

[4]

SAMPLE

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13. A delivery driver keeps a record of the number of miles per gallon (mpg) their car achieves each day.

These are the results for week 2:

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
30	38	22	37	40	32	39

This is a summary of the results for week 1:

Week 1	
Mean mpg	32
Range	16

The delivery driver claims:

a) *'on average, the car achieved more miles per gallon in week 2 than in week 1'*
b) *'the miles per gallon achieved in week 2 were more consistent than in week 1.'*

Are **each** of the driver's claims correct?

Give reasons for your answers.

Show your working.

Show any workings here:

Week 2 Mean = (30+38+22+37+40+32+39) ÷ 7 = 238 ÷ 7 = 34

Week 2 Range = 40 - 22 = 18 (Week 1 Range was 12)

Comment regarding claim a)

Claim 1: Yes, mean mpg is higher (34 > 32).

Claim 2: No, range is higher (18 > 12), so performance was less consistent. ✓ [4 marks]

Comment regarding claim b)

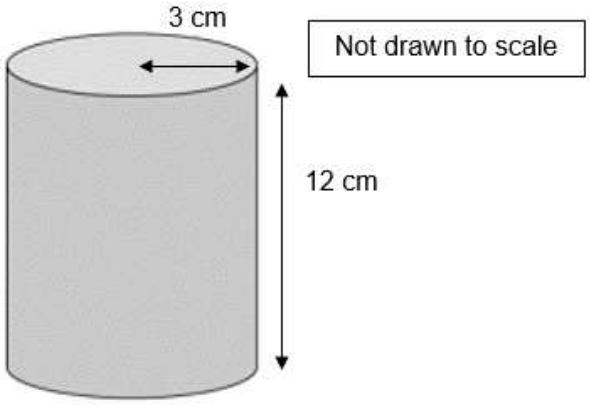
[4]

Questions continue on the following page

SAMPLE

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14. A nursery asks Cameron to make 50 cylinder-shaped toy blocks using this design:



When the blocks are made, Cameron needs to paint them using glow-in-the-dark paint.

Each tin has enough paint to cover 1150 square centimetres (cm^2).

How many tins of paint will Cameron need to buy to paint all 50 toy blocks?

Use $\pi = 3.14$

Surface Area = $2 \times (\pi \times 3^2) + (2 \times \pi \times 3 \times 12) = 56.52 + 226.08 = 282.6\ cm^2$

50 containers = 282.6 × 50 = 14 130 cm^2

Tins needed = 14 130 ÷ 1150 = 12.28... tins --> Rounds up to 13 tins

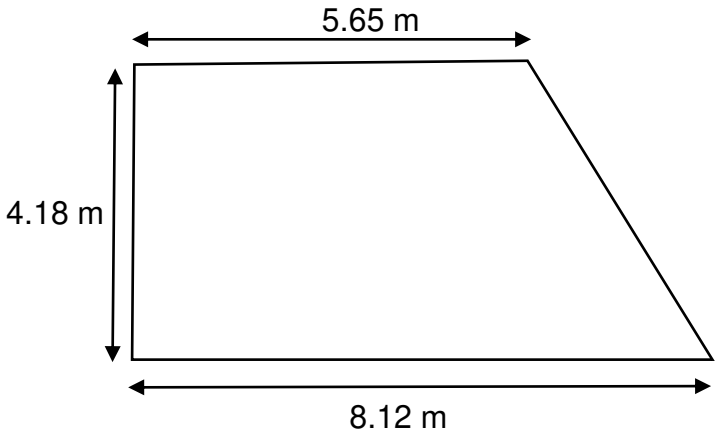
Answer	13 ✓ tins [5 marks]

[5]

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15. A family wants to cover this garden with grass.



Not drawn to scale

They will buy grass as turf in square metres (m^2).

The family finds these prices online:

Cost per square metre (m^2) Must buy a whole number of square metres					
£5.00	£3.80	£4.50	£5.00	£4.25	£6.00

The family wants to pay the same amount as the modal cost.

How much will it cost to cover the entire garden in turf?

Show your working.

Triangle Area = 0.5 × 2.47 × 4.18 = 5.1623 m^2

Rectangle Area = 5.65 × 4.18 = 23.617 m^2 --> Total Area = 28.78 m^2 --> 29 m^2

Modal cost of seed = £5.00/m^2. Total Cost = 29 m^2 × £5.00 = £145

Answer £ 145 ✓ [6 marks]

[6]

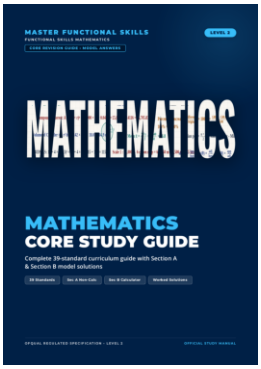
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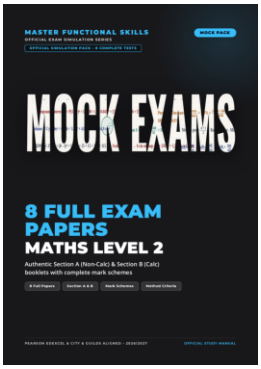


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